



LECTURE 2

Databases, DBMS & SQL

Databases, DBMS & SQL

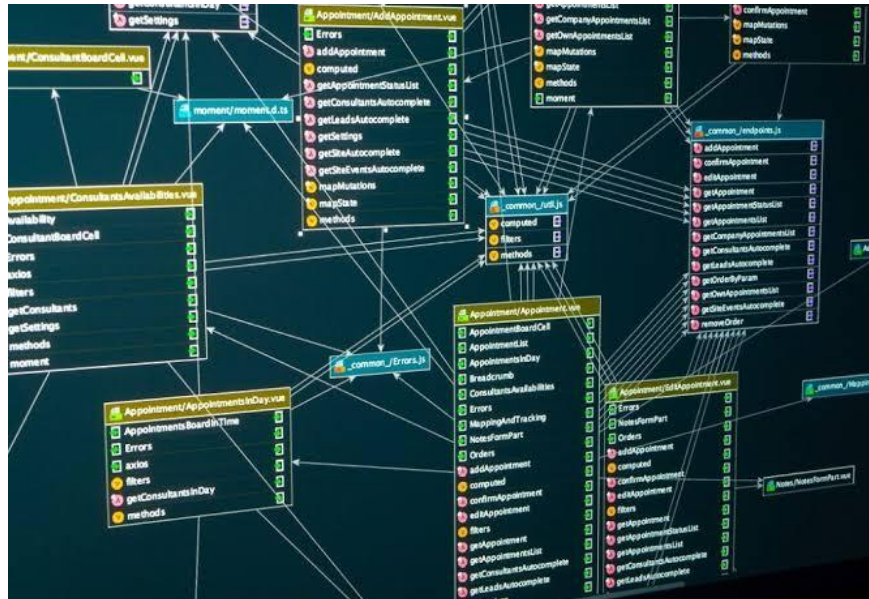
Most companies store their information in databases.

They all use at least one Database Management System (DBMS), such as MySQL, PostgreSQL, Microsoft SQL Server, or Oracle, and most systems support SQL.

Regardless of which programming language is used to implement business processes in a company (Python, C, C++ etc.), SQL is still needed to retrieve the required information from databases.

Databases

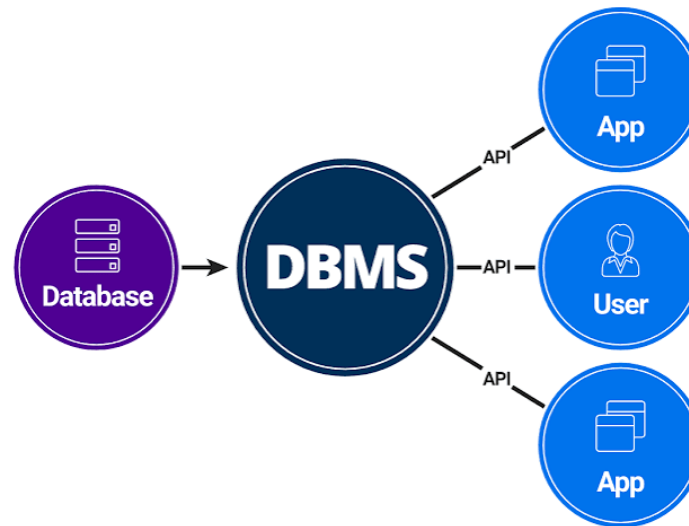
A database is a collection of information organized according to certain rules that define common principles for describing, storing, and manipulating data.



DBMS

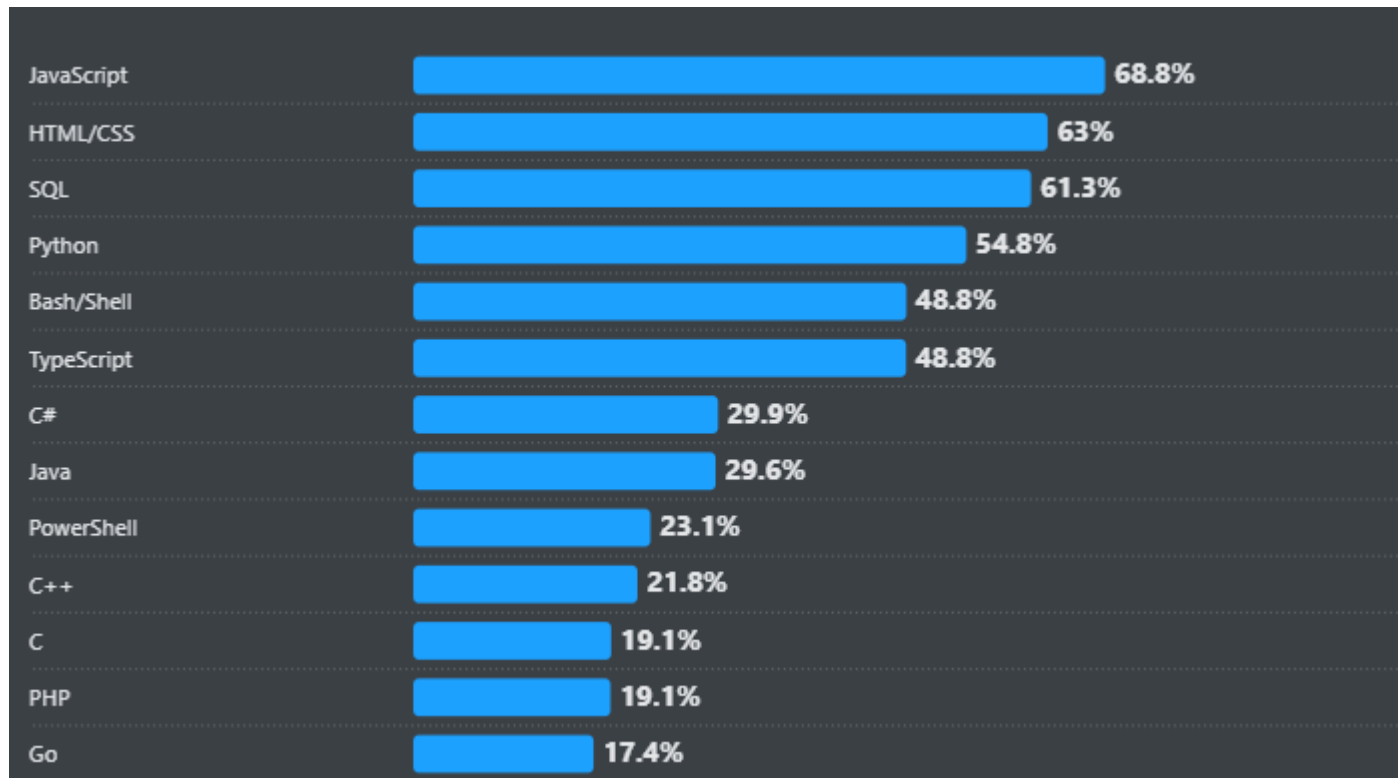
A Database Management System (DBMS) is a set of software tools that allows users to create a database and manipulate data (insert, update, delete, and retrieve it).

The system ensures security, reliable storage, and data integrity, and also provides tools for database administration.



SQL (Structured Query Language)

SQL is a Structured Query Language used to process information stored in a relational database.



Database Interaction Workflow



DBMS Rank

DBMS Ranking as of January 2026 (according to the DB-Engines*)

429 systems in ranking, January 2026

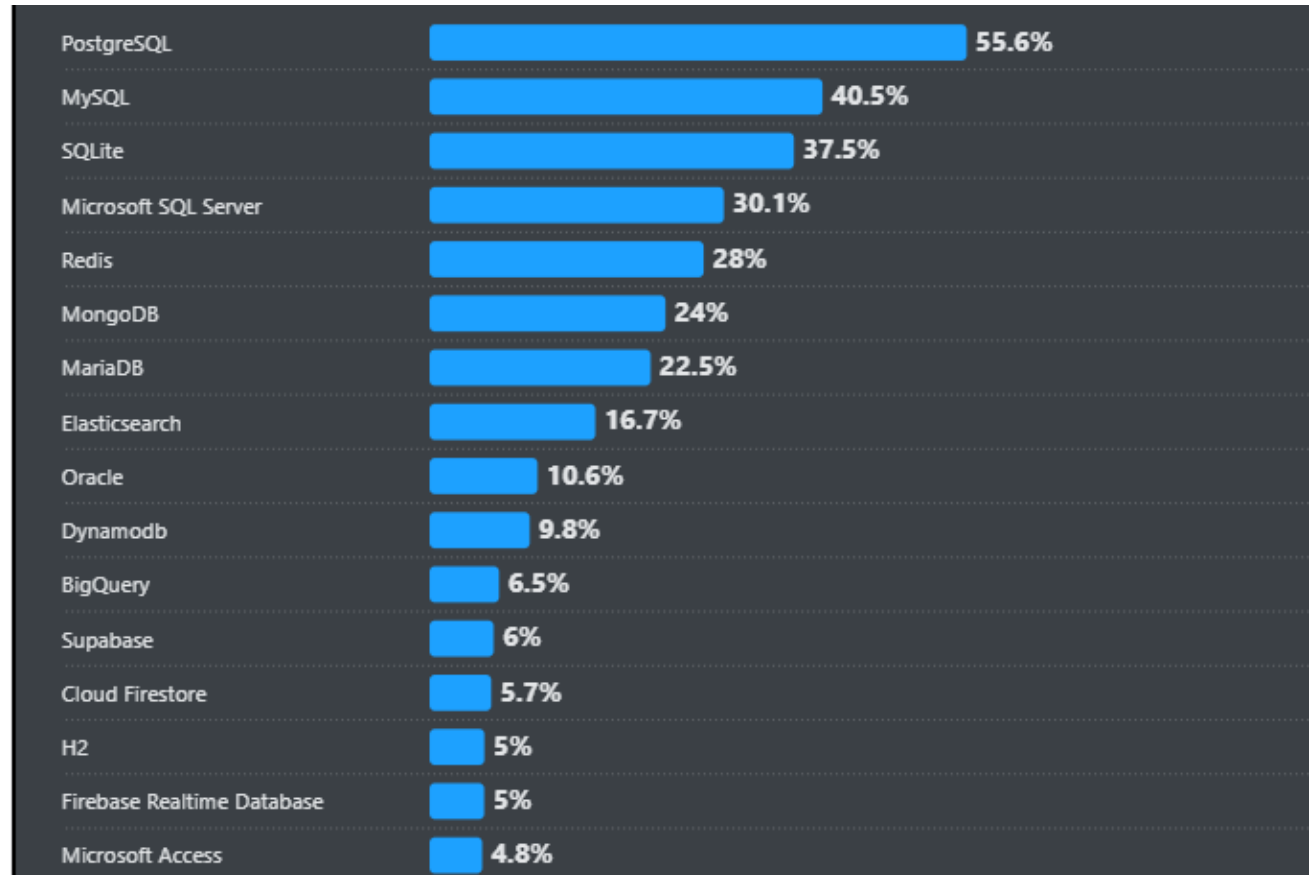
Rank			DBMS	Database Model	Score		
Jan 2026	Dec 2025	Jan 2025			Jan 2026	Dec 2025	Jan 2025
1.	1.	1.	Oracle	Relational, Multi-model 	1237.34	+2.94	-21.42
2.	2.	2.	MySQL	Relational, Multi-model 	867.52	-0.97	-130.63
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model 	706.26	-16.26	-92.30
4.	4.	4.	PostgreSQL	Relational, Multi-model 	666.27	+6.84	+2.86
5.	5.	5.	MongoDB	Multi-model 	376.74	+4.46	-25.77
6.	6.	6.	Snowflake	Relational	207.79	+5.34	+53.89
7.	7.	7.	Redis	Key-value, Multi-model 	144.16	+1.68	-9.20
8.	8.	 13.	Databricks	Multi-model 	141.55	+3.53	+53.70
9.	9.	9.	IBM Db2	Relational, Multi-model 	112.72	-3.04	-10.25
10.	10.	 8.	Elasticsearch	Multi-model 	107.15	-2.68	-27.78
11.	 12.	11.	Apache Cassandra	Wide column, Multi-model 	100.84	-0.71	+1.65
12.	 11.	 10.	SQLite	Relational	100.61	-2.01	-6.08
13.	13.	 14.	MariaDB 	Relational, Multi-model 	87.72	+0.13	+2.14
14.	14.	 15.	Splunk	Search engine	75.31	-0.58	-7.78
15.	 16.	 16.	Microsoft Azure SQL Database	Relational, Multi-model 	74.19	-1.03	+0.41

* Experts involved in this project track the global popularity of 428 database management systems. The ranking methodology resembles the TIOBE index for programming languages and takes into account search engine query popularity, the number of search results, the volume of discussions on popular forums and social networks, the number of job openings at recruitment agencies, and mentions in user profiles.

DBMS Rank



Each year Developer Survey explores the tools and technologies developers are currently using and the ones they want to use.



The significant growth in usage for Redis (+8%) highlights its growing importance. As applications become more complex, the need for high-speed, in-memory caching and data structures has made Redis an essential part of the modern tech stack.

SQL (Structure Query Language)

Main advantages of SQL:

- accuracy – no need to store redundant data;
- flexibility – even the most complex queries are easy to execute;
- scalability – multiple users can work with a single database;
- security – access to data in the tables is restricted to specific users.

SQL HISTORY

The SQL language appeared after the creation of relational algebra.

The goal was to create a simple non-procedural language that could be used by any user, even without programming skills.

Its prototype was developed in the late 1970s at IBM Research.

It was implemented in IBM's first relational DBMS, System R.

SQL HISTORY

The query language was developed by Donald D. Chamberlin and Ray Boyce.



SQL HISTORY

Later, SQL was used in many commercial DBMSs and gradually became a standard for data manipulation languages in relational databases.

Originally, the language was called SEQUEL (Structured English Query Language), but later the word “English” was dropped from the name.

SQL is designed to allow users to formulate queries to relational databases in a convenient and understandable way.

SQL, capabilities

Initially, SQL allowed the following operations:

- creating a database;
- creating a new table in the database;
- adding new records to the table;
- modifying records;
- deleting records;
- modifying table structures;
- retrieving records from one or more tables (based on a specified condition).

SQL, capabilities

New language capabilities:

- window functions;
- indexes;
- temporary tables and views;
- stored procedures and functions;
- triggers;
- and much more.

SQL, standardization

SQL standardization began almost simultaneously with the emergence of its first commercial implementations.

In 1982, the database committee of the American National Standards Institute (ANSI) was tasked with developing a specification for a standard relational database language.

SQL, standardization

Year	Name	Another name	Changes
1986	SQL-86	SQL-86	The first version of the standard, adopted by ANSI and approved by ISO in 1987.
1989	SQL-89	FIPS 127-1	A slightly modified version of the previous standard
1992	SQL-92	SQL-2, FIPS 127-2	Significant changes (ISO 9075): the Entry Level of the SQL-92 standard was adopted as FIPS 127-2.
1999	SQL:1999	SQL-3	Support has been added for regular expressions, recursive queries, triggers, basic procedural extensions, non-scalar data types, and certain object-oriented features.
2003	SQL:2003		Extensions for working with XML data, window functions, sequence generators, and data types based on them have been introduced.
2006	SQL:2006		Functionality for working with XML data has been significantly expanded; it is now possible to use SQL and XQuery together in queries.
2008	SQL:2008		Window function capabilities have been improved, and certain ambiguities in the SQL:2003 standard have been resolved.
...			
2023	SQL:2023		Built-in support for graph queries has been added. JSON handling has been improved, and useful functions for validating and processing JSON documents have been added.

SQL as a programming language

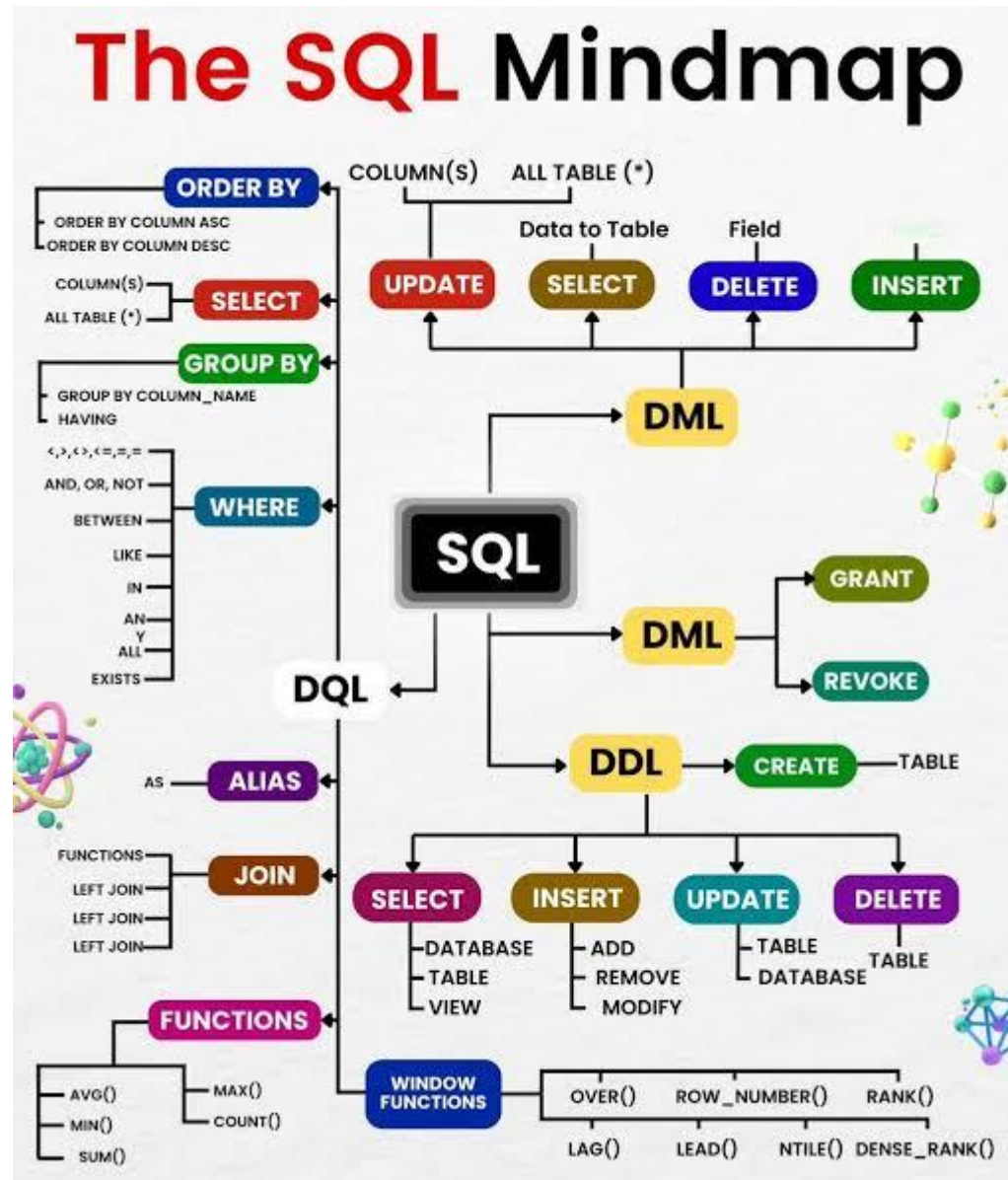
The SQL language is a collection of operators, statements, and computational functions.

SQL is classified as a declarative programming language.

SQL cannot be fully classified as a traditional programming language because it does not contain traditional constructs that control program execution (loops, etc.), type declaration operators, and so on.

SQL contains only a set of standard operators for accessing data stored in a database.

SQL, purpose



SQL, purpose

SQL operators are divided into:

- Operators for Data Definition Language, DDL:
 - **CREATE** creates a database object (database, table, view, etc.),
 - **ALTER** modifies an object,
 - **DROP** deletes an object;
- Operators for Data Manipulation Language, DML:
 - **SELECT** - selects data satisfying the specified conditions,
 - **INSERT** adds new data,
 - **UPDATE** modifies existing data,
 - **DELETE** deletes data.

SQL, purpose

- Operators for Data Control Language, DCL - manages user access rights to databases and objects:
 - **GRANT** - grants permissions to perform actions (e.g., read, write) to a user or role.
 - **REVOKE** - revokes previously granted access rights.
- Data Query Language, DQL - the primary and sole key operator is **SELECT**, but contains numerous keywords and subqueries:
 - **FROM** - specifies the table or tables - the data sources.
 - **WHERE** - specifies a row filtering condition.
 - **GROUP BY** - groups rows by the specified fields for aggregate functions.
 - **HAVING** - filters groups after applying GROUP BY.
 - **ORDER BY** - sorts the result set.

SQL, using

«SQL is so widespread that it permeates everything here. It is as if SQL syntax persists across time and space. Everything uses SQL or SQL derivatives».

Andy Pavlo, professor of Carnegie

SQL is used in their work by:

- analysts;
- product marketers;
- developers;
- testers;
- executives and managers.

SQL and Data Analytics

Data analysts are a kind of experimenters who have tools for combining data streams from different sources and for formulating and testing hypotheses.

This is where they need databases and a language that allows them to formulate precise queries. Obtaining the correct source data is an art that greatly increases the likelihood that in the behavior of the objects being studied cause-and-effect relationships will be identified. It is the quality of data that makes it possible to choose the most appropriate hypotheses.

SQL and Product Marketer

A product marketer analyzes and optimizes advertising to make it effective.

They develop recommendations for improving performance indicators.

Based on data, they help make decisions about a product to make it attractive to customers and develop a strategy for many months ahead.

They create forecasts and optimize advertising campaigns and the target budget.

SQL and Developers

Developers are people who create web content and software, administer websites and databases, and perform related tasks.

For example, web developers use SQL to access databases whose information is displayed on website or web application pages, as well as to manipulate this data.

The most popular toolset among web developers (backend developers) is a programming language (JAVA, Python) + SQL.

SQL and Testers

A tester analyzes and tests software and web content according to test scenarios and looks for potential errors. If errors are found, the tester writes a special report that programmers use to fix these errors.

A tester needs knowledge of programming fundamentals, the ability to work with databases using SQL, and basic knowledge of program language.

Resume

SQL is a Structured Query Language used to process information stored in a relational database.

It is used for:

- ✓ creating tables and databases;
- ✓ modifying table structures;
- ✓ populating tables with data;
- ✓ retrieving data from tables;
- ✓ updating and deleting data from tables;
- ✓ much more.

SQL:

- ✓ is universal and has a clearly defined structure;
- ✓ enables fast interaction with databases, even with large volumes of data;
- ✓ is used by almost all relational databases (MySQL, PostgreSQL, Oracle, etc.).

Quiz Questions

1. What is SQL used for when working with relational databases?
2. What are the main advantages of SQL? Explain what they entail.
3. Key stages in the evolution of SQL (presented as a diagram).
4. What primary operations did SQL originally allow users to perform?

Provide at least three examples.

5. Into which main groups are SQL statements categorized? What are DDL and DML, and what is their purpose?
6. What is the purpose of the CREATE, ALTER, and DROP statements?
7. What is the purpose of the SELECT, INSERT, UPDATE, and DELETE statements? How do the operations of data retrieval, insertion, modification, and deletion differ?
8. In which IT professions is SQL used? Explain why a data analyst, a developer, and a tester might need SQL.